

Initial outcomes in the Multicenter Study of Epilepsy Surgery

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Abstract—Objective: To obtain prospective data regarding seizures, anxiety, depression, and quality of life (QOL) outcomes after resective epilepsy surgery. **Methods:** The authors characterized resective epilepsy surgery patients prospectively at yearly intervals for seizure outcome, QOL, anxiety, and depression, using standardized instruments and patient interviews. **Results:** Of 396 patients who underwent resective surgical procedures, 355 were followed for at least 1 year. Of these, 75% achieved a 1-year remission at some time during follow-up; patients with medial temporal (77%) were more likely than neocortical resections (56%) to achieve remission ($p = 0.01$). Relapse occurred in 59 (22%) patients who remitted, more often in medial temporal (24%) than neocortical (4%) resected patients ($p = 0.02$). QOL, anxiety, and depression all improved dramatically within 3 months after surgery ($p < 0.0001$), with no significant difference based on seizure outcome. After 3 months, QOL in seizure-free patients further improved gradually, and patients with seizures showed gradual declines. By 12 and 24 months, overall QOL and its epilepsy-targeted and physical health domains were significantly different in the two outcome groups. (Anxiety and depression scores also gradually diverged, with improvements in seizure-free and declines in continued seizure groups, but differences were not significant.) **Conclusion:** Resective surgery for treatment of epilepsy significantly reduces seizures, most strikingly after medial temporal resection (77% 1 year remission) compared to neocortical resection (56% 1 year remission). Resective epilepsy surgery has a gradual but lasting effect on QOL, but minimal effects on anxiety and depression. Longer follow-up will be essential to determine ultimate seizure, QOL, and psychiatric outcomes of epilepsy surgery.

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Roughly 30% of adults with localization-related epilepsy have medically refractory seizures. In such patients, resection of the region of brain causing seizures is one approach to treatment, because it offers the potential for cure of seizures and successful psychosocial rehabilitation.^{1,2} Despite increasing use of surgery, the basis for determining which patients are the best candidates (i.e., most likely to achieve seizure freedom) is still largely based on results of retrospective analyses in small numbers of dissimilar patients treated with widely variable protocols over many decades. Even with established efficacy for seizures at 1 year after temporal resections according to a recently published randomized trial,³ the outcome of resective epilepsy surgery in other brain regions, over long-term follow-up, and incorporating effects on quality of life (QOL) and psychiatric status remains incompletely defined.^{4–7} We report the initial results of a multicenter, prospective study of resective epilepsy surgery, including nearly 400 patients evaluated and treated under a uniform protocol and followed for 1 to 5 years after surgery, with

standardized measures of seizure outcome, QOL, and psychiatric status.

Methods. Patients aged 12 years and older were enrolled at one of seven participating centers (Yale University, New York University, Montefiore Medical Center, Columbia University, University of Rochester, Jefferson Medical College, and the Minnesota Comprehensive Epilepsy Program). The inclusion criteria were as follows: 1) recurrent partial seizures (with or without secondary generalized seizures) for at least 2 years at an average rate of one per month totaling at least 20 in that 2-year period; 2) failure of at least two first-line antiepileptic drugs to control seizures; 3) no prior epilepsy surgery. Patients were invited to participate during their initial presurgical evaluation period. An appropriate consent procedure was followed with all prospective participants, and written informed consent was obtained from all who agreed to participate in the study. All procedures were approved by an Institutional Review Board. Subsequently, evaluation for surgery followed a standardized protocol consistent with current clinical practice, and differed for mesial temporal and neocortical epilepsies. Evaluation included a series of studies to establish localization of the epileptogenic region, seeking concordant electrophysiologic (scalp interictal and ictal EEG), structural (MRI), and functional (neuropsychological assessment, SPECT, PET) evidence. Lack of sufficient positive EEG localization with structural imaging confirmation, presence of discordant localization from imaging and EEG studies, or localization in proximity to

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critical cortical areas led to implanted electrode studies with combined subdural and intracerebral electrodes for additional information. Some specific applications of intracranial studies included all patients with suspected neocortical regions of epileptogenesis who lacked a structural lesion, to distinguish lateral from medial temporal seizure onset, or to clarify the presence or extent of unilateral versus bilateral independent medial temporal seizure onset.

Prior to surgery, patients (with or without help from family members) responded to a structured in-person interview administered by trained research associates at each center. The interview obtained information regarding seizure history, clinical and demographic characteristics, family information, employment, and daily activities. All participants who were at least 17 years of age at initial evaluation also completed standardized measures of QOL (Quality of Life in Epilepsy [QOLIE]-89), depression (Beck Depression Inventory [BDI]), and anxiety (Beck Anxiety Inventory [BAI]).⁸⁻¹² (Different measures, including the Child Health Questionnaire [CHQ] and the Child Behavior Checklist, were administered to enrollees under age 17 years because the QOLIE-89, BAI, and BDI are not appropriate for children and young adolescents.) Since discharge after surgery, this cohort has been closely followed. Phone calls are made every 3 months to ascertain seizure outcomes. The QOLIE-89, BAI, and BDI are re-administered at 3 and 12 months after surgery and once a year thereafter for 5 years.

Remission and relapse. For analysis of 1-year remission, and subsequent relapse after remission, we limited the cohort to those followed at least 1 year after surgery, the minimum needed to achieve this outcome. Remission was considered a period of at least 1 year seizure-free with or without isolated auras. A relapse was defined as the occurrence, after attaining remission, of any clinical seizure including simple or complex partial or generalized tonic-clonic (but not an isolated aura). (In this context, the term aura is used as commonly employed in practice and by patients to signify a subjective warning of impending seizure. In some instances, it may represent a simple partial seizure, but a strict relationship to epileptiform discharge is not always demonstrable. Because most epilepsy surgery literature makes the distinction in outcome categorization, we preserved this usage to provide results that are comparable to published outcome reports.)

Quality of life, depression, and anxiety. Because the cohort is still being actively followed, we limited analyses of QOL, depression, and anxiety to those observed through 2 years after surgery, and limited the analytic group to those followed at least 2 years. Patients missing a presurgical instrument or more than one of the first three postsurgical administrations of an instrument were omitted from analysis of that particular instrument. These analyses were also restricted to patients enrolled at age 17 and older, as patients who were younger at the time of enrollment were administered different, developmentally appropriate measures of these constructs.

QOL outcomes included the overall QOLIE-89 score as well as each of the four domain scores (cognitive distress, physical health, mental health, and the epilepsy-targeted domain). The QOLIE-89 is a self-report, generic and disease specific, validated and reliable, overall health-related QOL measure for the epilepsy population, developed in 1995. It incorporates the Short Form-36 as a generic core, supplemented by additional items targeting health-related QOL for adults with epilepsy. Underlying health-related QOL dimensions contributing to the epilepsy-targeted domain include work/driving/social function, medication effects, seizure worry, and health distress. Support for the construct validity and responsiveness of the QOLIE-89 has been reported.^{8,9} In analyses presented here, all scores were standardized to a mean of 50 with a SD of 10.⁸ Higher scores indicate better QOL.

The Beck Depression and Anxiety Inventories are well known, widely applied, reliable, and validated general measures of depression and anxiety that are scaled from 0 to 65 with higher scores reflecting more severe symptoms, and defined ranges for none, mild, moderate, and severe depression or anxiety. Recommended cutoffs were used to identify patients who had no or mild depression or anxiety (scores < 16) or moderate to severe depression or anxiety (scores ≥ 16).¹⁰⁻¹²

To determine the relationship between postsurgical seizure outcome and possible changes in postsurgical QOL, depression, and anxiety, seizure outcome was classified into one of four cate-

Table Characteristics of study sample (n = 355)

Characteristics	Value	Range
Male, n (%)	174 (49)	
Age at onset, y, mean (SD)	14.6 (11.6)	0–56
Age at study entry, y, mean (SD)	36.6 (11.3)	14–66
Duration of epilepsy, y, mean (SD)	22.0 (12.6)	1–64
Mesial temporal resection, n (%)	312 (87.9)	

gories: completely seizure-free for the first 2 years following surgery (excellent), seizure-free during the second year after surgery but with one or more seizures during the first year (good), 1-year seizure-free at some point during the first 2 years but with seizure recurrence during the second year (fair), never 1-year seizure-free during the first 2 years after surgery (poor).

Type of surgery. Surgery was designated as medial temporal versus neocortical (any other neocortical resection including temporal neocortex).

Analysis. Simple bivariate analyses were performed with χ^2 , *t*-tests, correlations, and one-way analysis of variance as appropriate for the data.

Longitudinal multiple regression analysis¹³ was used to examine postsurgical changes in the QOLIE scores (overall and the four domain scores). In these analyses, the presurgical score was treated as a covariate. Logistic regression¹⁴ was used to examine changes in depression and anxiety after surgery. The product limit method was used to estimate the probability of attaining 1 year remission after surgery and the probability of having a first relapse. Results are displayed as Kaplan-Meier curves.¹⁵ For remission, time was counted from the day of discharge from the hospital and only seizures occurring after discharge have been considered (i.e., immediate postoperative seizures are excluded). Patients who did not achieve remission were censored as of their last date of follow-up. Time for relapse began with the day the patient achieved a 1-year remission. Patients who did not relapse were censored as of their last date of follow-up.

Results. A total of 788 patients were initially identified to the study. Of those, 169 were fully evaluated but did not have surgery, and 396 had surgery. The remaining 223 patients were excluded for a variety of reasons: 6 died, 54 were later found not to meet eligibility criteria, and 163 withdrew, were lost to follow-up, or had incomplete surgical information.

Of the 396 surgical patients in this cohort to date, 355 patients have been followed at least 1 year after surgery (range 1 to 5 years), representing 1,085 total person-years of follow-up. Simple descriptive characteristics of the sample are provided in the table.

Remission, relapse, and mortality. Of the 355 patients followed at least 1 year, 264 (75%) experienced a 1-year remission at some time during follow-up. The probability of attaining a 1-year remission at 1, 2, 3, and 4 years after surgery was 0.53 (0.47, 0.57), 0.70 (0.65, 0.75), 0.75 (0.60, 0.80), and 0.79 (0.73, 0.84) (figure 1A). More patients with medial temporal resections achieved remission at some point during follow-up compared to those with neocortical resections (77% versus 56%, *p* = 0.01, figure 1A).

Relapse occurred in 59 (22%) patients who remitted. The probability of relapsing by 1 or 2 years after achieving a 1-year remission was 0.16 (0.12, 0.21) and 0.25 (0.18, 0.30). More patients in the medial temporal lobe group relapsed compared to those with neocortical resections (24% versus 4%, *p* = 0.02, figure 1B).

A "reason" for relapse was given in 80% of patients. Reduction (38%) or discontinuation (8%) of medications



Figure 1. Cumulative probability of (A) remission after resective surgery and (B) relapse after attaining remission in patients with medial temporal resections (solid line) and neocortical resections (dotted line).

was the reported cause in the largest number of patients, with noncompliance in 17%, and other identified causes in 17% (mostly illness or sleep deprivation).

Eight deaths occurred in this surgically treated cohort, all after temporal lobe resections. Four occurred in the first year of follow-up, two in the second year, and the remainder 3 or more years after surgery. Three of the eight were in remission at the time of death.

Quality of life, depression, and anxiety. Completeness of baseline and follow-up data. Of the 396 subjects, 14 were adolescents for whom a different QOL instrument was used and whose data were therefore not eligible for this analysis; there was not a sufficient number for a separate analysis. Full QOL data were available on 362 of the 382 adults (94%). Of the 254 patients who had baseline QOLIE measurements and were followed at least 2 years after surgery, 56 (22%) had two and 175 (69%) had three of the first three postsurgical QOLIE-89s. Thus, 23 (9%) of those who had been followed at least 2 years were excluded from analysis of the QOLIE due to missing postsurgical data.

The presurgical BAI were available for 360 and the presurgical BDI for 358. Of those followed at least 2 years, 25 had to be excluded due to missing postsurgical BAI and BDI. For the 251 patients included in the analysis of the postsurgical BAI, 50 (20%) had only two and 176 (70%) three of the first three postsurgical measures. Of the 249 included in the analysis of the postsurgical BDI, 47 (19%) had only two and 177 (71%) three of the first three postsurgical measures.

Comparisons of those excluded and included. To determine whether potential bias may have entered the analyses due to exclusion of some patients, comparisons between patients who were and were not included were performed. For the QOLIE as well as the Beck indices, there were generally no substantial or significant differences between included versus excluded patients with respect to presurgical scores, age at onset of epilepsy, age at surgical evaluation, education, and ultimate postsurgical seizure outcome. In addition, for those who had presurgical scores, there were no differences in these scores between those who did versus did not complete all the postsurgical QOLIE and Beck questionnaires. The only difference found between excluded and included patients was that, of those excluded from analysis of each measure, 73% were men, versus 43% men among those included ($p = 0.03$).

Analysis of presurgical scores. Because this was not a randomized trial, we explored any potential differences in baseline measures of QOL, depression, and anxiety as a

function of patient characteristics as well as postsurgical seizure outcome to help ensure that any postsurgical differences associated with seizure outcome were indeed postsurgical and not pre-existing or due to other factors. None of the presurgical QOLIE scores or Beck measures was correlated with postsurgical seizure outcome. Some differences in presurgical levels were observed in association with level of education and sex. Those with higher levels of education tended to have scores that reflected slightly worse QOL and higher levels of depression. Women also tended to have scores reflecting worse QOL and higher levels of anxiety prior to surgery. In a simple correlation analysis, increasing or higher age at surgical evaluation was negatively correlated with all the QOLIE measures and positively correlated with higher depression scores. We had 90% or greater power to detect differences of 0.5 standard deviations for these comparisons, so our ability to detect baseline differences in those who became seizure free and those who did not was excellent.

Overall postsurgical changes in QOLIE. Changes in QOLIE-89 (overall and domains) over time for the entire group are shown in figure 2. Scores rose substantially from before surgery to 3 months after surgery ($p < 0.0001$). Beyond that point, there were minimal additional gains, and most changes were not significant. The one exception was in the epilepsy-targeted domain, which continued to show improvement over the first year. Accounting for the

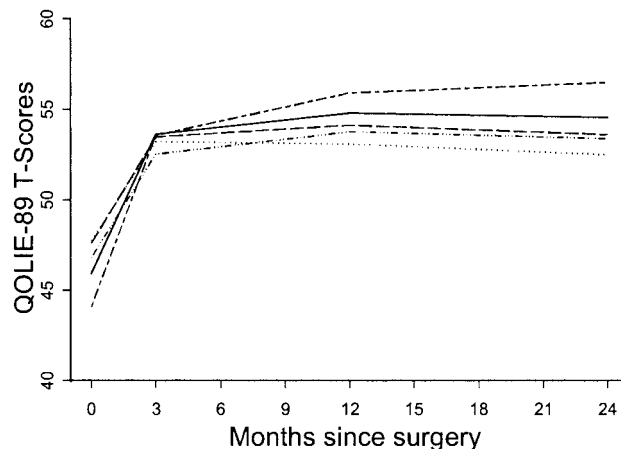


Figure 2. Mean Quality of Life in Epilepsy (QOLIE-89) t-scores overall (solid line) and for cognitive distress (dotted line), physical health (— · —), mental health (— · — · —), and epilepsy targeted (— · —) domain scores.

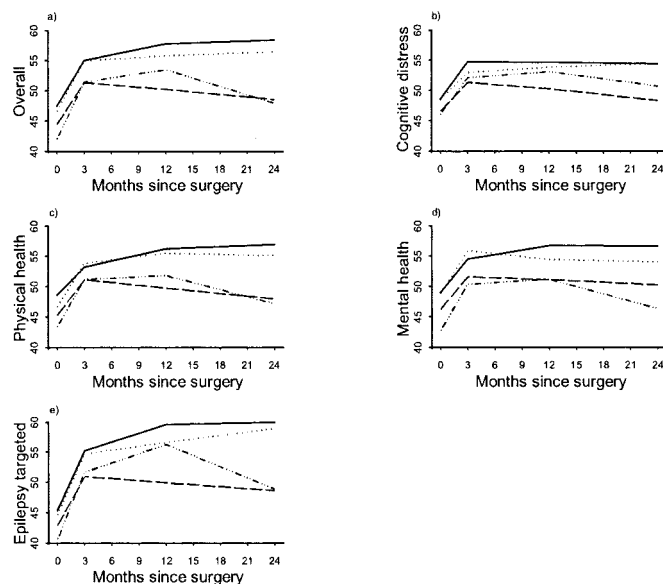


Figure 3. Change in Quality of Life in Epilepsy-89 (A) overall, (B) cognitive distress, (C) physical health, (D) mental health, and (E) epilepsy-targeted domain t-scores over time since surgery by seizure outcome: seizure-free since discharge ($n = 111$) (solid line), some seizures in first year, seizure-free at 2 years ($n = 42$) (dotted line), seizure-free after surgery with recurrences in second year ($n = 23$) (— · —), and never 1 year seizure-free during first 2 years of follow-up ($n = 78$) (-----).

correlation between scores for an individual over time, the increase was significant between 3 and 12 months ($p < 0.0001$), but not from 12 to 24 months ($p = 0.12$).

Seizure outcome and postsurgical QOLIE. All four seizure outcome groups experienced some improvement in overall QOLIE scores at 3 months (figure 3A). QOLIE scores were higher at 3 months after surgery for those who were ultimately completely seizure-free, and/or 1 year seizure-free after 2 years, compared with the other two seizure outcome groups, but the difference was not significant. This trend continued over time, with the worst two groups losing some of the initial postsurgical gains.

In longitudinal regression analysis, changes in scores over time as a function of seizure outcome were examined. Compared to those who never became seizure-free (worst outcome group), the completely seizure-free (best) group had higher scores on the overall QOLIE ($p = 0.01$) and on the physical ($p = 0.002$) and epilepsy-targeted ($p = 0.001$) domain scores. The only difference between the seizure-free at 2 years (good) group and the worst group was in the epilepsy-targeted domain score ($p = 0.02$). No significant differences were found between the worst and the fair (remission but relapsed before 2 years) groups. Effects were most dramatic in the physical health and epilepsy-targeted domain scores, which accounted for most of the overall score changes (figure 3, B through E).

Focusing on the postoperative time only, the effects of seizure outcome on QOLIE over time were generally modest. For example, for the epilepsy-targeted domain, where the greatest effects were observed, the average score rose from 55.2 3 months after surgery to 59.5 at 12 months and

61.3 at 24 months in the best (continuously seizure-free) seizure outcome group.

Examination of QOLIE scores over time revealed a close parallel between responses of the excellent and good seizure outcome groups, and between the responses of the fair and worst seizure outcome groups. In many instances, seizure-free in the second year was almost the same as seizure-free continuously, while relapse in the second year had as much negative impact on QOL as never seizure-free. Seizure status in the most recent year before assessment, rather than cumulative seizure experience, seems most allied to QOL outcome.

Overall postsurgical changes in BAI and BDI. Both indices decrease (reflecting less anxiety and depression) right after surgery (3 months). Some of the initial gains were lost for depression by 12 months, while the anxiety measure decreased slightly (improved) over time. The mean scores prior to surgery and at 3, 12, and 24 months after surgery were 10.6, 6.5, 6.2, and 6.1 for anxiety and 10.4, 6.8, 7.4, and 6.9 for depression.

Seizure outcome and postsurgical anxiety and depression. The group in remission since surgery showed the greatest improvements over time. As for QOL, the excellent and good outcomes, and the fair and poor groups, behaved almost identically.

Because the depression (BDI) and anxiety (BAI) scores did not sufficiently follow the statistical assumptions needed for valid application of linear regression algorithms, our longitudinal analyses focused on a dichotomized version of the scales. For the anxiety measure, there were 26%, 10%, 10%, and 13% of patients labeled as anxious prior to surgery and at 2, 12, and 24 months. For depression, the figures are 24%, 10%, 13%, and 13%. In longitudinal logistic regression analyses in which presurgical depression or anxiety was treated as a covariate, few differences were seen among the seizure outcome groups at 3 months after surgery, and no differences beyond that point. In all analyses, sex, education, age at onset, and age at surgical evaluation were entered into the model as covariates. After adjustment for presurgical depression or anxiety, none was significantly associated with postsurgical depression or anxiety.

Discussion. Previous work invariably showed significant improvement in seizures with resective temporal and extratemporal epilepsy surgery, in children and adults.¹⁶⁻³⁰ Improvement was established by actuarial analyses, analyses at fixed follow-up, and analyses at latest follow-up. Furthermore, reports came to remarkably similar estimates of the likelihood of seizure freedom after resective surgery, amounting to 65 to 80% of lesional or temporal lobe resected patients, and 45 to 55% of extratemporal or neocortical resected patients.¹⁶⁻³⁰ Our early (1 year) seizure remission rate was comparable to those retrospective analyses, and to the recently reported 1-year randomized temporal lobe surgery trial results.³ Previous reports were based on smaller populations across many years with different types of evaluation and variable surgical approaches. Nevertheless, the seizure-free rate (measured by 1 year remission, exclusive of auras) seems to be a robust outcome.

A proportion of our patients continued to reach 1 year seizure-freedom with successive follow-up, particularly in the medial temporal group. Similarly, 22% of patients with 1 year remission relapsed after achieving a 1 year remission, and those were also usually medial temporal patients. The latter observation has been made before, but not in as large or contemporary a group, or prospectively.³¹⁻³⁴ Extratemporal, but not temporal patients, were the group found more likely to achieve later 1 year remission.³² One possible explanation for this difference is our categorization of the temporal group as only medial temporal resection, while others included both temporal neocortical and medial temporal resection in the "temporal" group. Those neocortical temporal patients may have diluted previous reports.

The 3-month improvement in almost all domains and overall QOL scores in all patients regardless of seizure outcome is remarkable. A similar phenomenon was observed in the randomized temporal lobe epilepsy surgery trial, for the surgical group as a whole.³ Several potential explanations exist for this observation. One is the timing or setting of the preoperative baseline QOL measurement. Measurement of QOL in patients with epilepsy during the anxiety-ridden preoperative interval might adversely impact QOL scores, as well as indices for anxiety and depression, which then improve (or actually recover) after surgery. Another possibility is that improved seizure control, albeit imperfect, may be sufficient to lead to improvements in reported QOL. Alternatively, removal of the epileptogenic region might itself be the factor that improves QOL, regardless of whether it produces control or reduction of seizures. Attention to side effects, or more effective and less depressant anticonvulsant regimens, might be initiated during evaluation for surgery and result in improvement in QOL. The results are compatible with a low baseline that "recovers" by 3 months, or with an influence of factors other than seizures or surgery on QOL and psychiatric status after epilepsy surgery.

Improvement in QOL continues after the 3 month spike (although at a much slower rate) in the best outcome groups. On the other hand, patients who are never seizure-free begin to decline immediately after 3 months, and relapsing patients begin to decline later. By 24 months, both groups with recurrent seizures are nearly back to the (relatively low) baseline values. In seizure-free patients, improvement occurs predominantly in the epilepsy-targeted domain of the QOLIE-89, not in more generic aspects. Although the QOLIE-89 directly and indirectly targets other aspects of QOL like employment, schooling, and education, we believe 2 years is insufficient time to adequately appreciate or analyze those types of changes.

Previous studies that examined QOL outcomes of resective epilepsy surgery, for the most part, did not follow patients prospectively, or measured QOL with nonstandardized instruments that incorporated only some domains.³⁵⁻³⁹ In one study that included a comparison group of nonsurgical patients, health-related

QOL was not significantly improved postoperatively,^{6,7} and in another, improvement took 2 years to be manifest, and then was seen in seizure-free individuals only.³⁹ Our results are compatible with those (limited) observations.

Depression and anxiety results in our population were notable for several reasons. First, the prospective measurement of these outcomes at several intervals of time after epilepsy surgery is an advance over previous studies. The literature that addresses psychiatric effects of epilepsy surgery is sparse^{6,20,40} with reports of increases or decreases in preoperative depressive symptomatology or appearance of new depressive psychoses postoperatively.^{6,20,40,41} Most postoperative affective disorders were transient and easily treated pharmacologically. To the extent that depression is highly correlated with QOL measures, one might predict the results.^{42,43} For both anxiety and depression, the same 3-month improvement (measured by Beck indices) in all patients (surgical, nonsurgical, temporal, other surgery) was seen. Later, as for QOL, slow changes paralleling seizure status occurred. We could invoke similar reasoning to explain these changes.

True and lasting improvements in QOL are likely to require a lengthy time to be manifest, and are the real goal. Longer term follow-up, in our contemporary, large, and uniformly studied prospective cohort, will be critical in defining the extent and overall impact of epilepsy surgery.

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